

STEP	MODEL CHANGE	WHAT CHANGED
01	Baseline refit	Refit the 2023 diagnostic configuration as the starting model.
02	Executable and INI	Update the MFCL executable and adopt INI format 1007.
03	Natural mortality	Fix the Lorenzen natural-mortality scaling parameter.
04	Length-weight	Update the bias-corrected BET length-weight parameters.
05	Spatial structure	Adopt five regions and 33 fisheries; remap reporting rates.
06	Weight-to-length data	Convert reweighted weight compositions to length.
07	Observed lengths	Add observed lengths where coverage exceeds weight samples.
08	2024 data update	Extend frequency and tagging data through 2024, except CAAL.
09	Size-data QC	Apply PH/ID and domestic mixed-gear size-data rules.
10	Regional CPUE	Introduce regional CPUE indices and abundance scaling.
11	CPUE uncertainty	Allow CPUE observation uncertainty to vary through time.
12	CPUE error SDs	Fix the five regional CPUE observation-error SDs at their MLE-based estimates.
13	New CAAL data	Add the new CAAL data with weight 0.75.
14a	Regional CAAL weights	Evaluate CAAL reweighting across all five regions.
14b	Sub-basin CAAL weights	Retain sub-basin CAAL reweighting for Regions 3 and 4.
15	Selectivity	Update fishery selectivity and add the weak F10 penalty.
16	Tag mixing	Apply tag-mixing periods estimated with a KS D-statistic cutoff of 0.20.
17	Tag reporting	Exclude reporting rates during pre-mixing periods.
18	Effort creep	Apply effort-creep adjustments to the regional CPUE indices.
19	Composition weighting	Apply Dirichlet-multinomial weighting (G8; $N_{\max} = 25$).
20	Tag overdispersion	Fix negative-binomial tag overdispersion at $\tau = 2$.
21	F33 selectivity penalty	Stabilize the data-limited F33 tail without forcing asymptotic selectivity.
22	Diagnostic model	Fix steepness at $h = 0.90$ and adopt the Diagnostic model.